TEMS Academy of Olympiad Math
Adhoc Algebra Problem Set

Md. Fuad Al Alam

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Problem 1. Evaluate the sum

$$\sum_{k=1}^{n} \frac{k}{2^k}$$

Problem 2. Let a, b, c, d and e be positive integers such that

$$abcde = a + b + c + d + e$$

Find the maximum possible value of $\max\{a, b, c, d, e\}$.

Problem 3. Evaluate the sum

$$\frac{3}{1! + 2! + 3!} + \frac{4}{4! + 5! + 6!} + \cdots + \frac{2027}{2025! + 2026! + 2027!}$$

Problem 4. Find all real numbers x for which

$$\frac{8^x + 27^x}{12^x + 18^x} = \frac{7}{6}$$

Problem 5. Evaluate the sum

$$\sum_{k=1}^{n} k!(k^2 + k + 1)$$

Problem 6. Find all real numbers x such that

$$(x^2 - 9x + 19)^{x^2 - 3x + 2} = 1$$

Problem 7. Let F_n be the nth Fibonacci number i.e. $F_0 = F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$. Evaluate the sum

$$\sum_{n=1}^{\infty} \frac{1}{F_{n-1} F_{n+1}}$$

Problem 8. Find all real numbers x such that

$$2^x + 3^x - 4^x + 6^x - 9^x = 1$$

Problem 9. Let a, b and c be distinct nonzero real numbers such that

$$a + \frac{1}{b} = b + \frac{1}{c} = c + \frac{1}{a}$$

Prove that $|abc| = 1$.

Problem 10. Find all real numbers x such that

$$10^x + 11^x + 12^x = 13^x + 14^x$$

Problem 11. Solve the system of equations

$$\begin{aligned}
(x_1 - x_2 + x_3)^2 &= x_2(x_4 + x_5 - x_2) \\
(x_2 - x_3 + x_4)^2 &= x_3(x_5 + x_1 - x_3) \\
(x_3 - x_4 + x_5)^2 &= x_4(x_1 + x_2 - x_4) \\
(x_4 - x_5 + x_1)^2 &= x_5(x_2 + x_3 - x_5) \\
(x_5 - x_1 + x_2)^2 &= x_1(x_3 + x_4 - x_1)
\end{aligned}$$

for real numbers x_1, x_2, x_3, x_4, x_5.

Problem 12. Let x and y be real numbers such that $x \neq 1$, $y \neq 1$ and

$$(x-1)^2 + (y-1)^2 = xy(x + y - 3) + 1$$

Show that $x + y = 1$.

Problem 13. Evaluate the sum

$$\sum_{k=0}^{\infty} \tan^{-1}\left(\frac{2}{k^2}\right)$$

Problem 14. Let x, y and z be complex numbers such that $x + y + z = 2$, $x^2 + y^2 + z^2 = 3$ and $xyz = 4$. Evaluate

$$\frac{1}{xy + z - 1} + \frac{1}{yz + x - 1} + \frac{1}{zx + y - 1}$$

Problem 15. Evaluate the sum

$$\binom{2027}{2} + \binom{2027}{5} + \binom{2027}{8} + \cdots + \binom{2027}{2027}$$